



life.augmented

**Create your time stamped
detection event
application launched by
the secure manager !**

Tomas Team

4

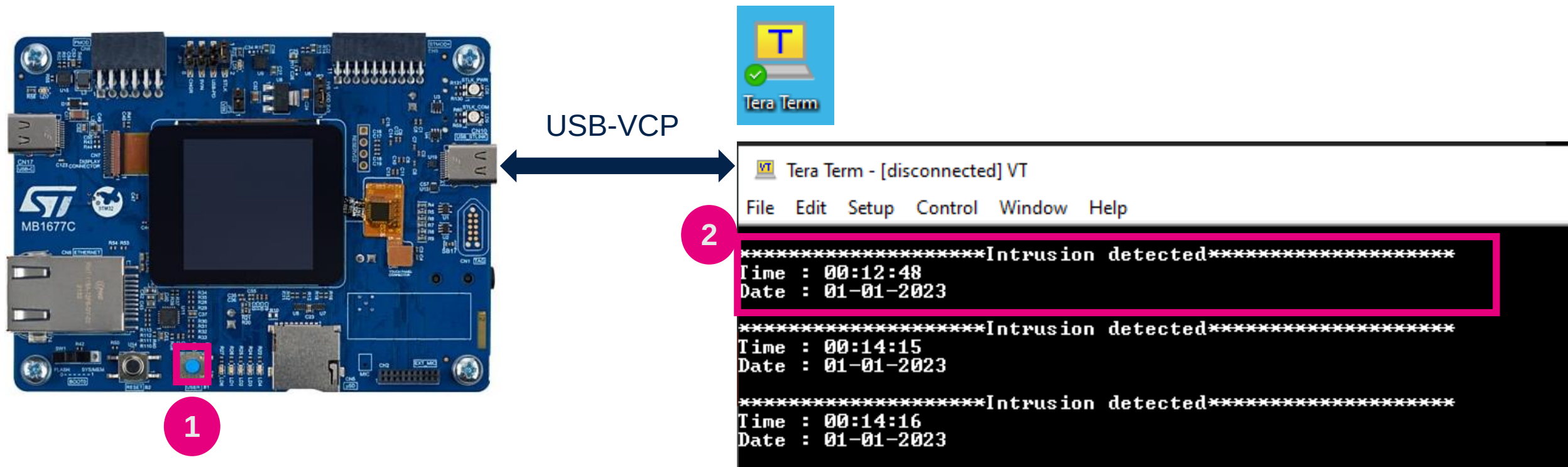
Create your timestamped event detection application launched by the secure manager !

Agenda Details

- Timing : 1 hour
- Purpose
 - Show integration of Secure Manager in ecosystem using CubeMX
- Process
 - Create a user application which will be launched by the secure manager
- Payoff
 - Easy to create an application in the secure manager context with CubeMX
 - Understand the boot scheme in the secure manager context



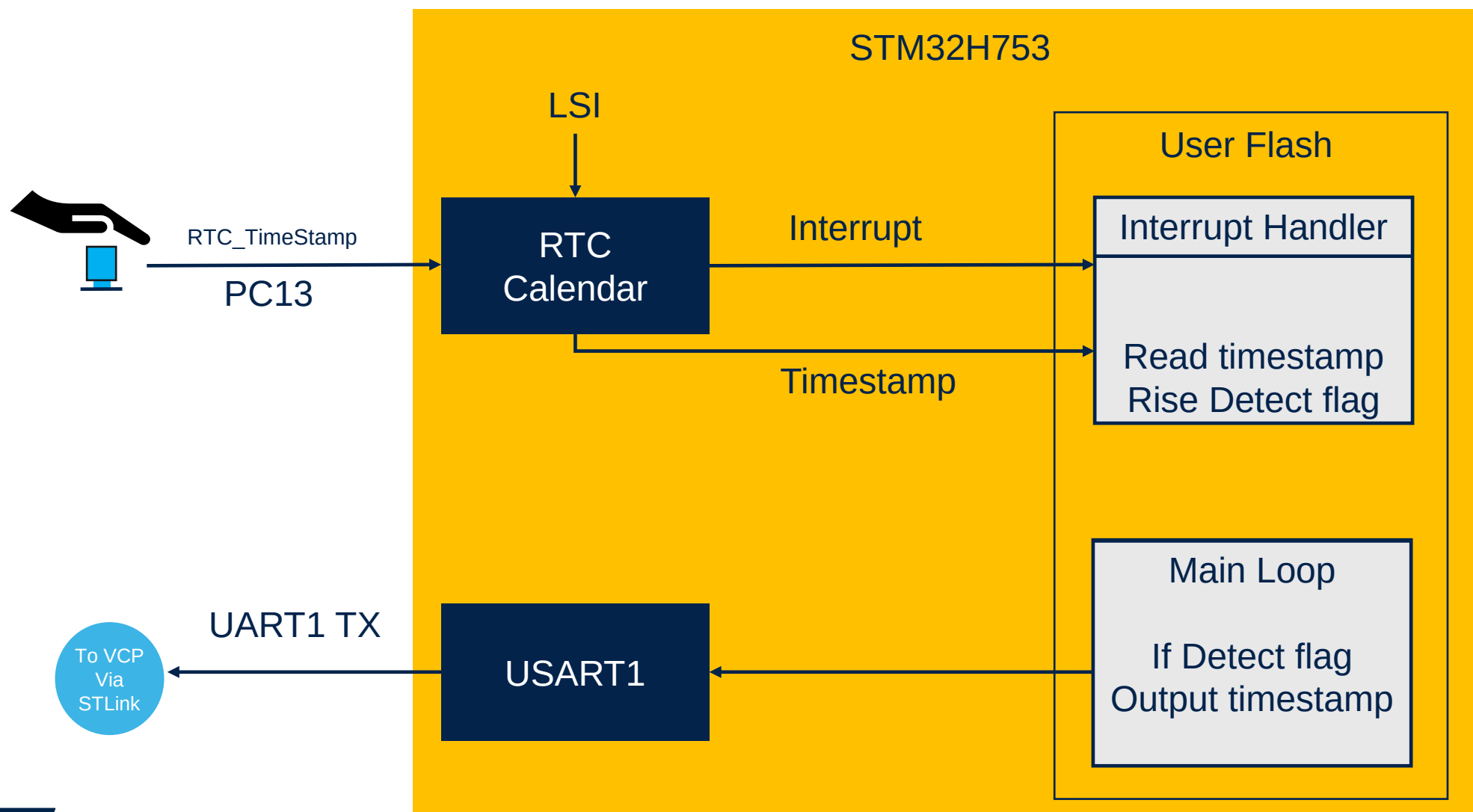
Application functional description



STM32 resources used : RTC Calendar / TimeStamp / USART1

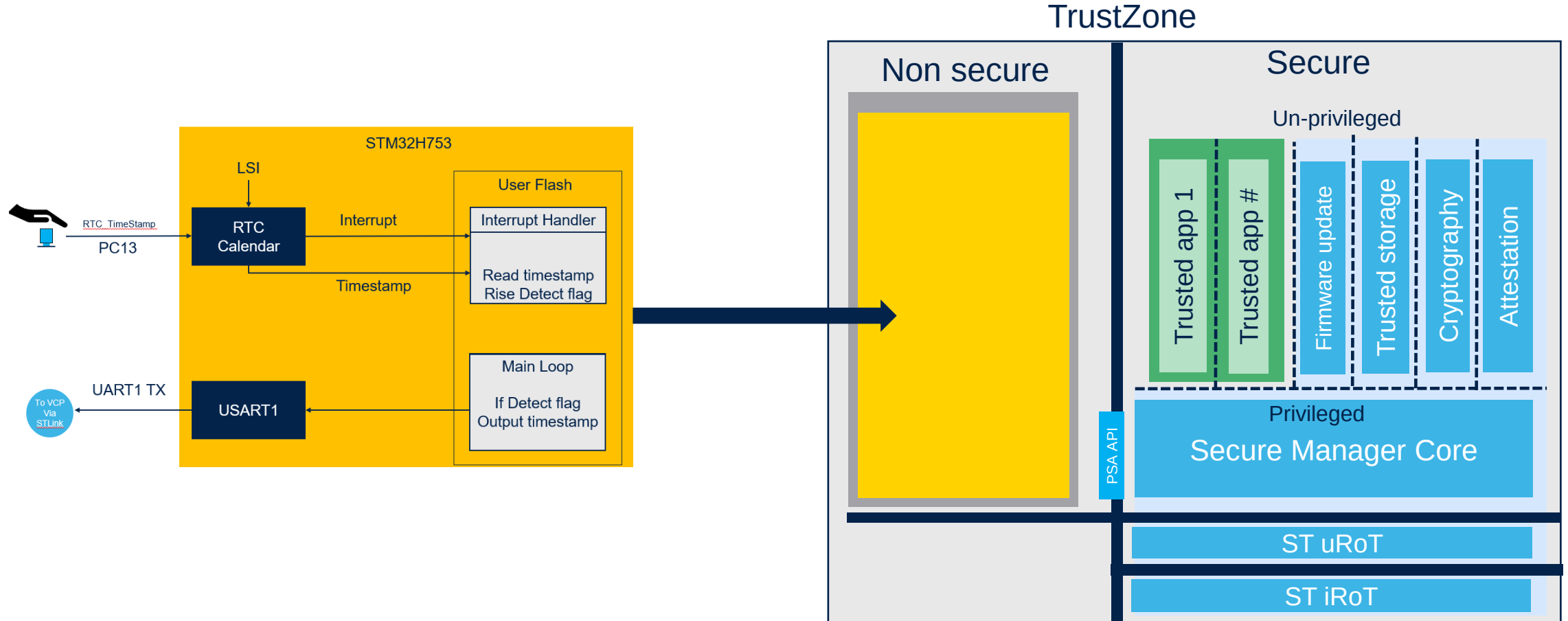


Principle of detection





User application





Launch CUBEMX, Select MCU selector

MX STM32CubeMX Untitled

File Window Help myST

Home

Existing Projects

Recent Opened Projects

Test.ioc	MX
Last modified date : 24/08/2023 13:56:05	
Test_again.ioc	MX
Last modified date : 23/08/2023 9:45:57	
Test_appli.ioc	MX
Last modified date : 22/08/2023 17:15:13	
CDC_Standalone.ioc	MX
Last modified date : 12/04/2023 10:48:53	
Last_test.ioc	MX
Last modified date : 03/08/2023 15:13:46	

Other Projects

New Project

I need to :

- 1 Start My project from MCU
ACCESS TO MCU SELECTOR
- Start My project from ST Board
ACCESS TO BOARD SELECTOR
- Start My project from Example
ACCESS TO EXAMPLE SELECTOR

Manage software installations

Check for STM32CubeMX and embedded software package...

CHECK FOR UPDATES

Install or remove embedded software packages

INSTALL / REMOVE

MX new in v6.9

Ability to launch user's scripts at pre- and post- project generation

Home > Project Manager > Code Generator > User Actions

About STM32 External Tools

[Data collection information notice](#)



Target is STM32H573IIK3Q

MX New Project from a MCU/MPU

MCU/MPU Selector | Board Selector | Example Selector | Cross Selector

MCU/MPU Filters

Commercial Part Number

PRODUCT INFO

Segment

Series

Line

Marketing Status

Price

Package

Core

Coprocessor

MEMORY

Flash = 2048 (kBytes)

2048

EEPROM = 0 (Bytes)

0

STM32H573II

STM32H5 Series

STM32H573IIK3Q

High-performance, Arm Cortex-M33 with TrustZone, MCU with 2-Mbyte Flash, 640-Kbyte RAM, 250 MHz CPU

Unit Price for 10kU (US\$) : 6.1102

MCUs/MPUs List: 14 items

	Commercial Part No	Part No	Reference	P...	Board	Flash	RAM
☆	STM32H573AII3Q	STM32H573AI	STM32H573AIIxQ	UF...		2048 kBytes	640 k...
☆	STM32H573AII6		STM32H573AIIx	UE		2048 kBytes	640 k...
☆	STM32H573IIK3Q		STM32H573IIKxQ	UF...	STM32H573I-DK	2048 kBytes	640 k...
☆	STM32H573IIK6		STM32H573IIKx	UF...		2048 kBytes	640 k...
☆	STM32H573IIK6TR	STM32H573II	STM32H573IIKx	UF...		2048 kBytes	640 k...
☆	STM32H573IIT3Q		STM32H573IITxQ	LQ...		2048 kBytes	640 k...
☆	STM32H573IIT6		STM32H573IITx	LQ...		2048 kBytes	640 k...
☆	STM32H573MIY3QTR	STM32H573MI	STM32H573MIYxQ	W...		2048 kBytes	640 k...
☆	STM32H573RIT6		STM32H573RITx	LQ...		2048 kBytes	640 k...
☆	STM32H573RIV6	STM32H573RI	STM32H573RIVx	VF...		2048 kBytes	640 k...
☆	STM32H573VIT3Q		STM32H573VITxQ	LQ...		2048 kBytes	640 k...
☆	STM32H573VIT6	STM32H573VI	STM32H573VITx	LQ...		2048 kBytes	640 k...
☆	STM32H573ZIT3Q		STM32H573ZITxQ	LQ...		2048 kBytes	640 k...
☆	STM32H573ZIT6	STM32H573ZI	STM32H573ZITx	LQ...		2048 kBytes	640 k...

Display similar items

Export

3

Start Project



TrustZone is activated

TrustZone feature available



Do you want to create a new project :

☐ without TrustZone activated ?

4

☒ with TrustZone activated ?

5

OK



Select name/path and IDE

Home > STM32H573IIKxQ > Untitled - Project Manager > **GENERATE CODE**

Pinout & Configuration | Clock Configuration | **Project Manager** | Tools

Project

Code Generator

Advanced Settings

Boot Path and Debug Authentication

Project Settings

Project Name: TimeStamped_Event_Detection

Project Location: C:\Training\STM32H5_Security_WS\Hands-on\ Browse

Project Structure: ☒ Secure Project ☒ Non Secure Project

Application Structure: Advanced ☐ Do not generate the main()

Toolchain Folder Location: C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\

Toolchain / IDE: STM32CubeIDE ☒ Generate Under Root

Linker Settings

	M33S	M33NS
Minimum Heap Size	0x200	0x200
Minimum Stack Size	0x400	0x400

Please avoid space character in path and in project !





Non-secure application only

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Project

Code Generator

Advanced Settings

Boot Path and Debug Authentication

Project Settings

Project Name

Project Location

Project Structure

Application Structure

Toolchain Folder Location

Toolchain / IDE

Linker Settings

Minimum Heap Size

Minimum Stack Size

Thread-safe Settings

CortexM33S

Enable multi-threaded support

Thread-safe Locking Strategy

TimeStamped_Event_Detection

C:\Training\STM32H5_Security_WS\Hands-on\

1

☐ Secure Project

☒ Non Secure Project

Advanced

Do not generate the main()

Warning: Project Structure

2

You have changed your project Structure.

- Only Non Secure Code will be generated and Secure Code won't be removed.

- If some Ips are secured they will be ignored. Please check from the System View Tab.

OK

Default – Mapping suitable strategy depending on RTOS selection.



Save the project

The screenshot shows the STM32 CubeMX software interface. The 'File' menu is open, and the 'Save Project' option is highlighted. A red circle with the number '1' is placed over the 'File' menu button, and a red circle with the number '2' is placed over the 'Save Project' option. A tooltip for 'Save Project' says 'Save the current Project'. The interface includes a top bar with 'Home', 'STM32H573IIIKxQ', and 'GENERATE CODE'. Below this are tabs for 'Pinout & Configuration', 'Clock Configuration', 'Project Manager', and 'Tools'. The left sidebar has 'Project', 'Code Generator', and 'Advanced Settings'. The main area shows project settings like 'TimeStamped_Event_Detection', 'C:\Training\STM32H5_Security_WS\Hands-on\', 'Secure Project' (unchecked), 'Non Secure Project' (checked), 'Advanced' (selected), 'C:\Training\STM32H5_Workshop\TimeStamped_Event_Detection\', and 'STM32CubeIDE' (selected) with 'Generate Under Root' checked.

- CubeMX IOC file is saved. Saving at this stage is mandatory to go further



Select boot path

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Project

Code Generator

Advanced Settings

Boot Path
and Debug Authentication

Debug Authentication

Configure

Path Selection

Select

No boot path selected (Nothing to configure)

Application (Default Boot Path)

Select bootpath

MX Boot Path

ST Define your boot Path

TZEN=1 'Only Non Secure' project

Boot Step#1

☐ Secure Application

☐ OEM immutable Root of Trust (OEM-iRoT)

☒ ST immutable Root of Trust (ST-iRoT)

Graphical Boot Representation

Non Secure	Secure
	ST-iRoT

STM32

BACK NEXT FINISH CANCEL

MX Boot Path

ST Define your boot Path

TZEN=1 'Only Non Secure' project

Boot Step#2

☐ Secure Application

☒ updatable Root of Trust (uRoT)

Graphical Boot Representation

Non Secure	Secure
	ST-uRoT
	ST-iRoT

STM32

BACK NEXT FINISH CANCEL

MX Boot Path

ST Define your boot Path

TZEN=1 'Only Non Secure' project

Boot Step#3

☐ Secure Application

☒ Secure Manager Non Secure Application

Graphical Boot Representation

Non Secure	Secure
Application	Secure Manager
	ST-uRoT
	ST-iRoT

STM32

BACK NEXT FINISH CANCEL

- When clicking “Finish”, ROT_Provisioning is added to the project



Boot path configuration done

STM32CubeMX File Window Help myST

Home > STM32H573I-KxQ > TimeStamped_Event_Detection.ioc - Project Manager

GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Project

Code Generator

Advanced Settings

Boot Path and Debug Authentication

1

Configure ☒ Generate DA Folder

Only with CubeMX 6.10.0 or upper

Boot Path Selection

Select ST_IROT_UROT_SECURE_MANAGER

☒ Use Default Reference Location

Reference Location C:\Users\lefrere\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\SM\ Browse

Config Files Location C:\Training\STM32H5_Workshop\TimeStamped_Event_Detection\ROT_Provisioning\SM

Linker

☒ Enable Linker File(s) Update

☐ Reinitialize Linker File(s)

Signature

☒ Sign Binary(ies)

Mapping

Start address 0x806E400

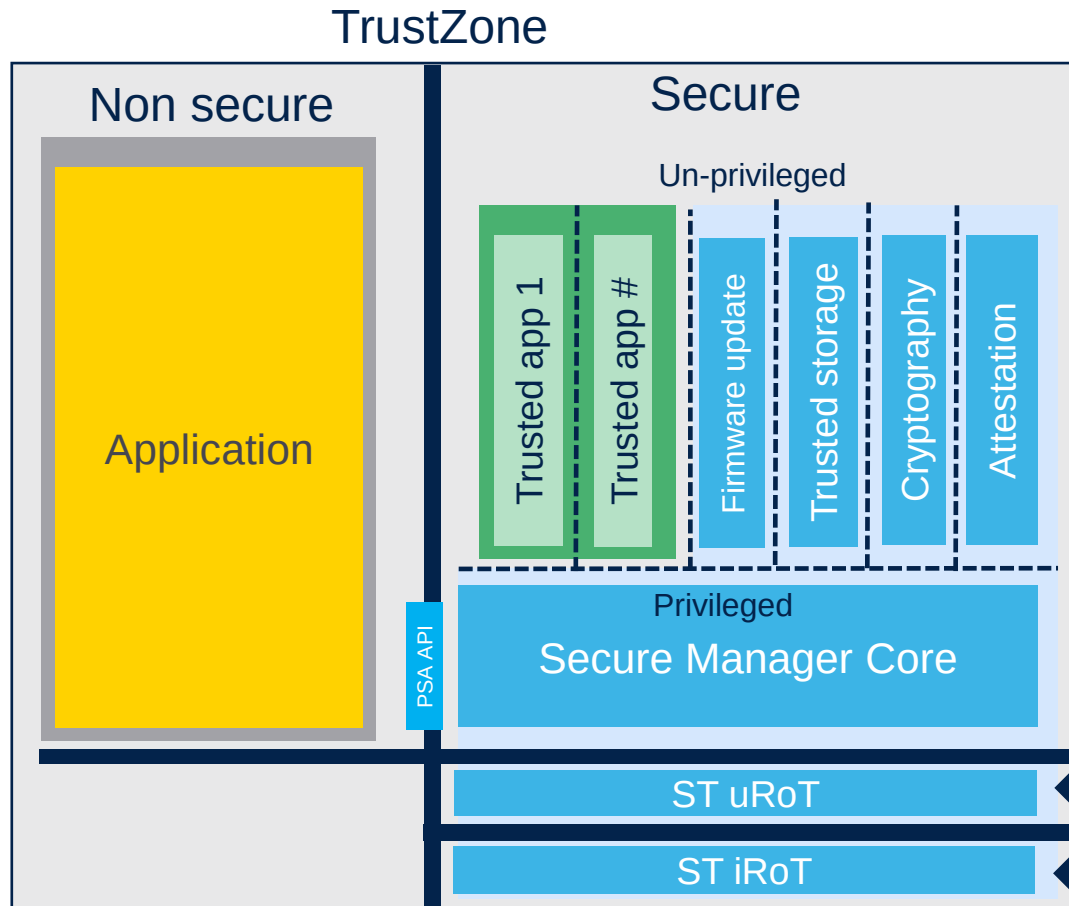
End address 0x80FF7FF

Boot Path Image

Non Secure	Secure
Application	Secure Manager
	ST-uRoT
	ST-IrOT
STM32	



Secure manager boot scheme



2 stage bootloader :

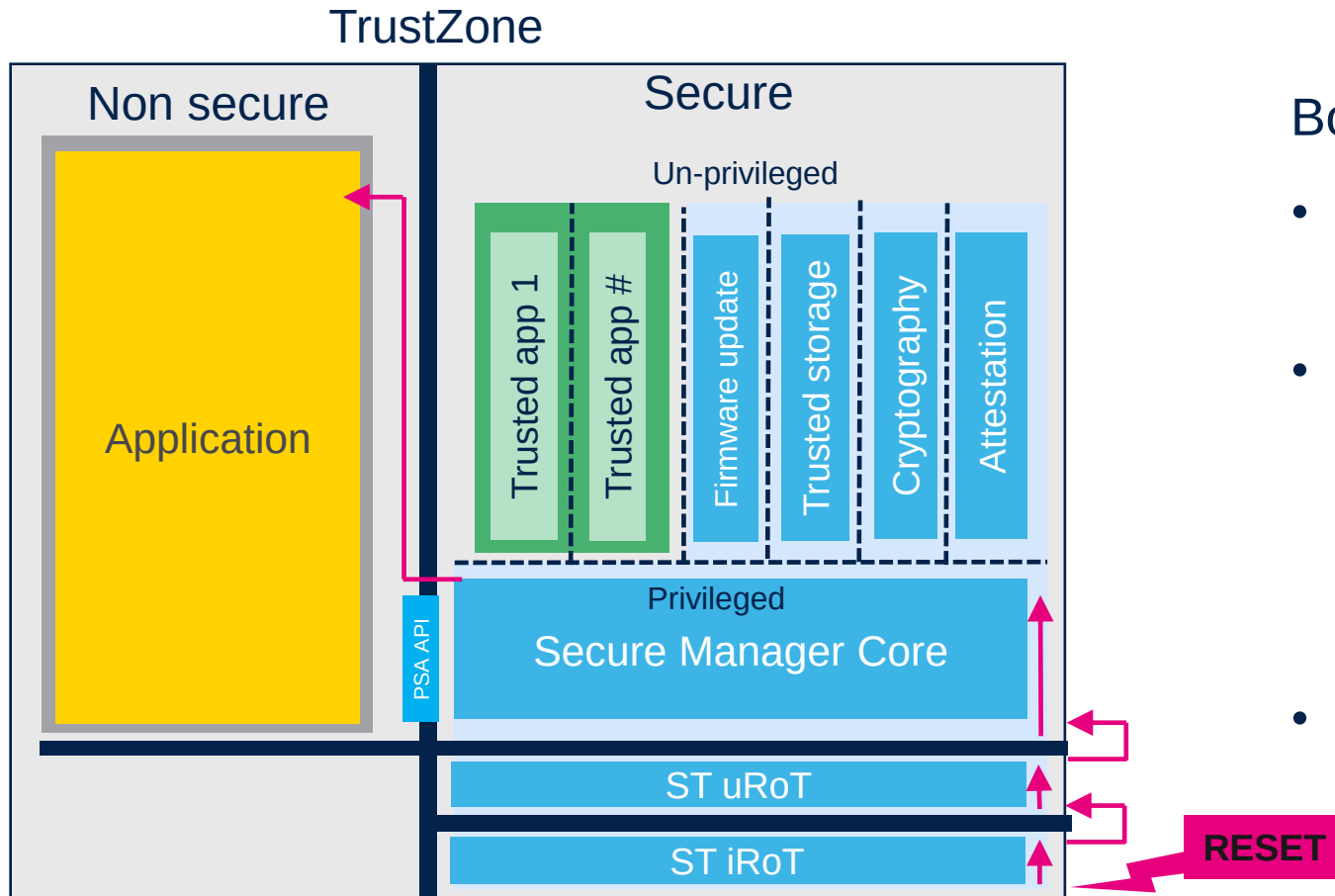
- STiROT could update STuROT
- STuROT could update secure manager / secure module / Application

STuROT : Software bootloader based on MCU boot with secure firmware update capability. Delivered by ST encrypted and signed.

STiROT : ROMed secure boot with secure firmware update capability



Secure manager boot scheme



Boot Process :

- Reset : STiROT check authenticity and integrity of STuROT and launch it if ok
- STuROT check authenticity and integrity of Secure manager (core, services and module), check authenticity and integrity of user application. If ok, launch Secure manager
- Secure manager will launch userapplication

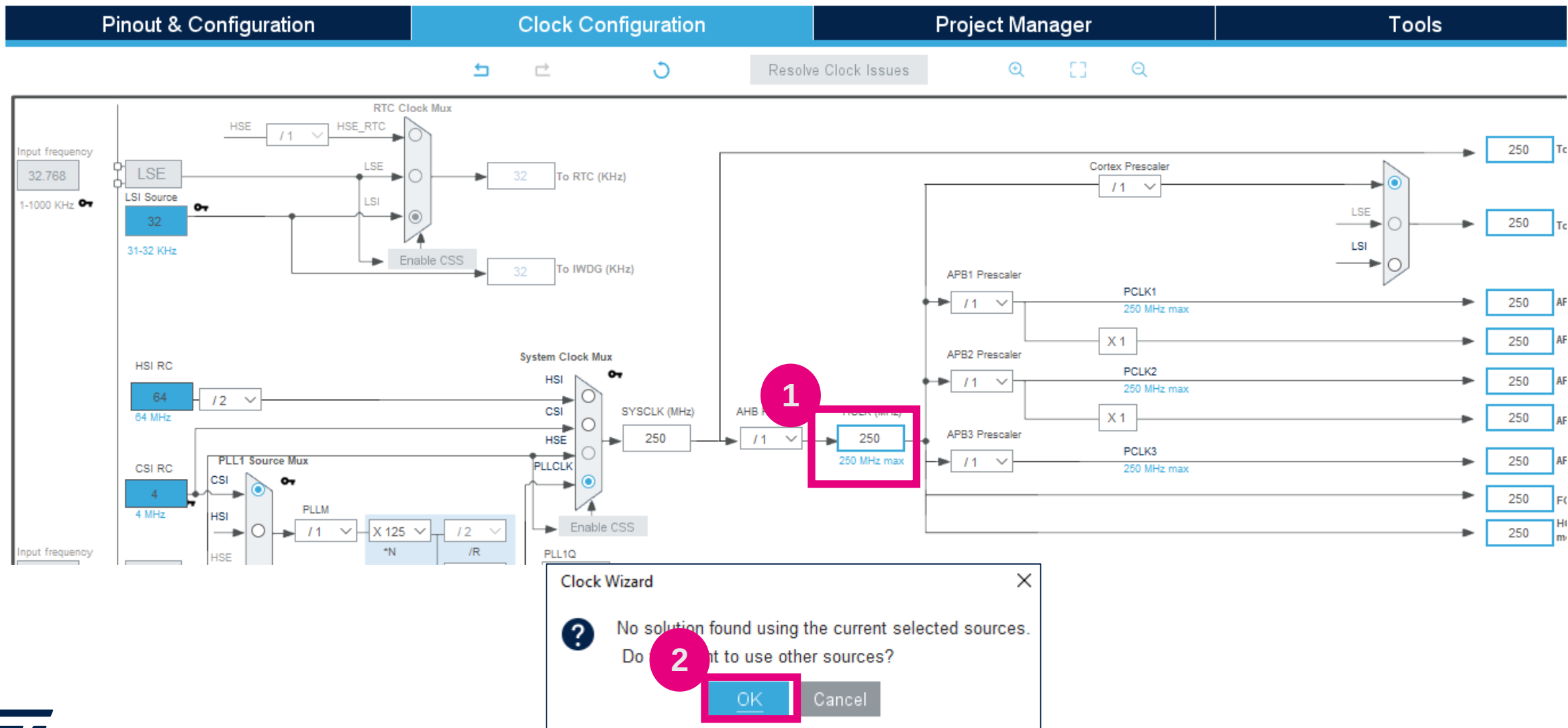


Boot path integrated in CubeMX

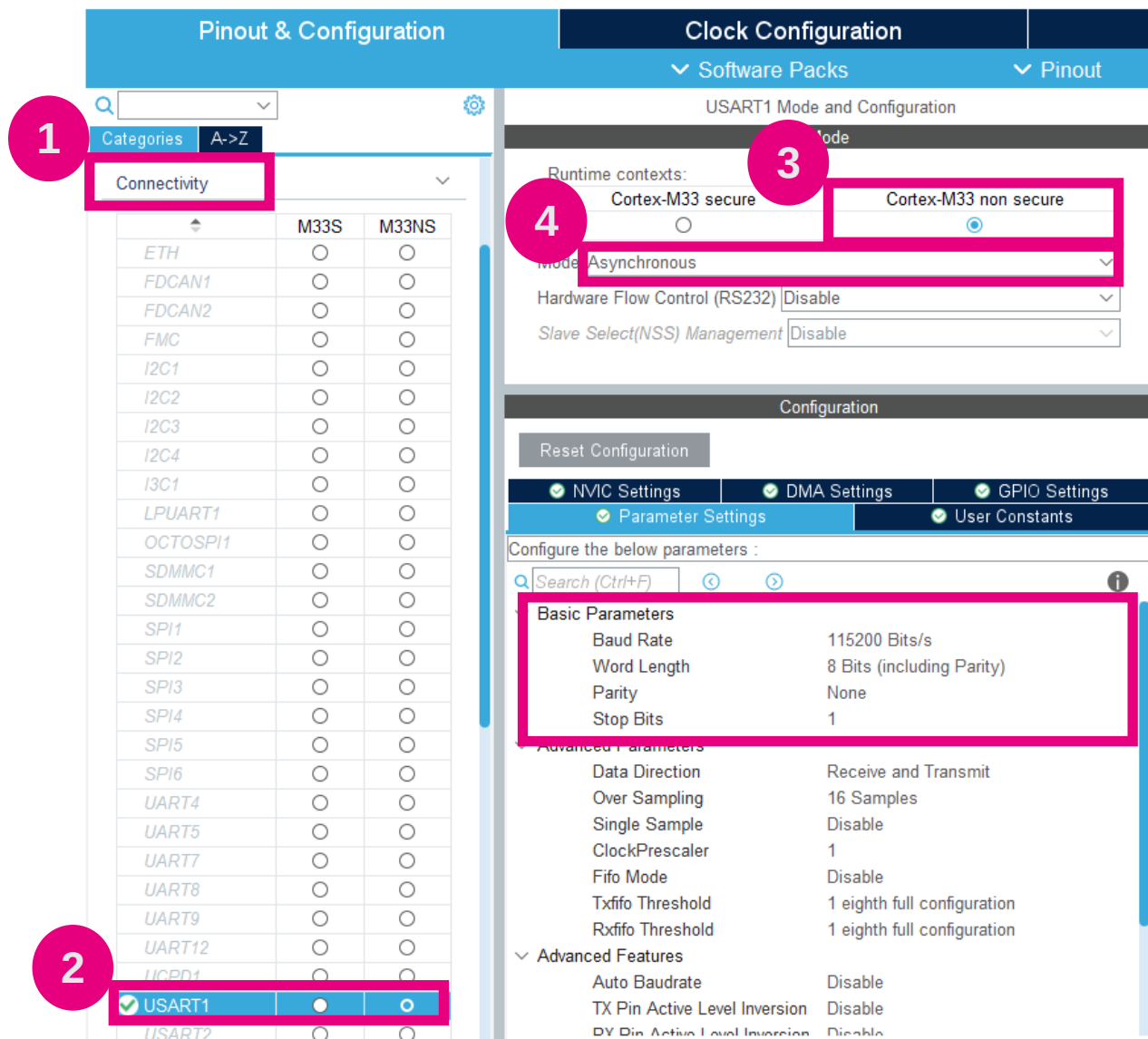
- New feature introduced with STM32H5 in STM32CubeMX
- Goal is to ease the usage of a Secure boot for secure manager and other use cases.
- This configuration in CubeMX allows :
 - To copy all the requested component in the project (STuROT/SecureManager)
 - To copy all the script that would be needed for an easy integration
 - To update the linker script of your application to be compliant with secure manager
- Many other bootpath will be supported in future version



Change the clocking



Configure the USART1



The screenshot shows the STM32CubeMX Pinout & Configuration window. The left sidebar (1) has the 'Connectivity' category selected. The main table (2) lists various peripherals, with 'USART1' selected. The right panel (3) shows the 'USART1 Mode and Configuration' settings. The 'Runtime contexts' section (4) shows 'Cortex-M33 non secure' selected. The 'Configuration' section shows the 'Parameter Settings' tab active, displaying the 'Basic Parameters' for USART1.

1 Categories: Connectivity

2 USART1

3 USART1 Mode and Configuration

4 Runtime contexts: Cortex-M33 secure, Cortex-M33 non secure

Configuration

Reset Configuration

NVIC Settings, DMA Settings, GPIO Settings, Parameter Settings, User Constants

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Baud Rate	115200 Bits/s
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

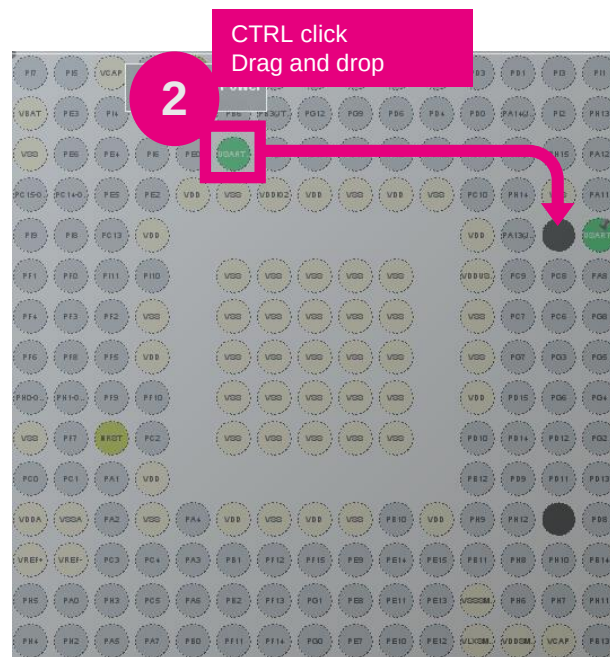
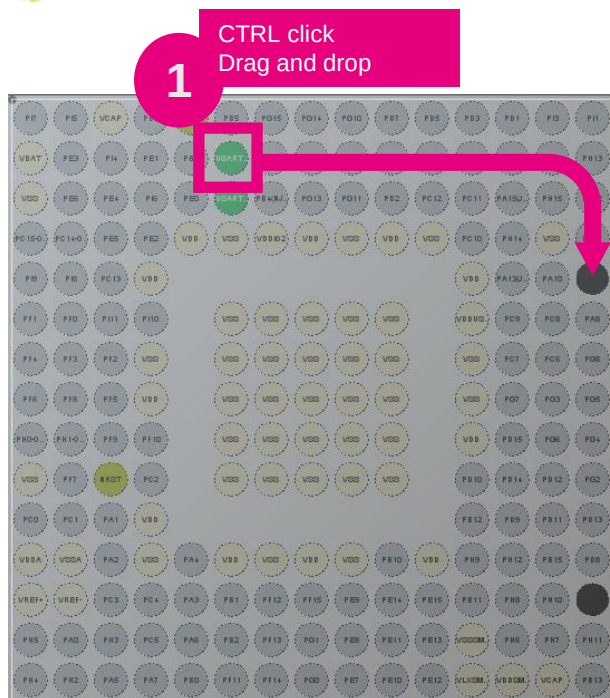
Advanced Features

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
PX Pin Active Level Inversion	Disable

Default configuration is OK



Remap USART1 pin to PA9/PA10



Configuration

Reset Configuration

✓ NVIC Settings ✓ DMA Settings ✓ GPIO Settings ✓ Parameter Settings ✓ User Constants

Search Signals

Search (Ctrl+F)

Pin ...	Signal o...	Pin Cont...	Pin Privl...	GPIO o...	GPIO m...	GPIO P...	Maxin
PA9	USART...	Cortex-...	n/a	n/a	Alternat...	No pull-...	Low
PA10	USART...	Cortex-...	n/a	n/a	Alternat...	No pull-...	Low

Default configuration is NOT OK

Need to remap both GPIOs of USART1 to PA9 and PA10

PA9 & PA10 are connected to embedded STLink

STLink will forward to USB VCP

Activate the RTC calendar

The screenshot shows the ST Studio IDE configuration interface for the RTC module. The interface is divided into two main sections: 'Pinout & Configuration' on the left and 'Clock Configuration' on the right. The 'Pinout & Configuration' section shows a list of components under 'System Core', 'Analog', and 'Timers'. The 'Timers' section is expanded, showing a table of timers (LPTIM1 to LPTIM6, RTC, TAMP, TIM1 to TIM17) with columns for M33S and M33NS. The 'RTC' component is highlighted with a red box and a red circle labeled '1'. The 'Clock Configuration' section shows the 'RTC Mode and Configuration' tab. The 'Mode' section has two radio buttons: 'Cortex-M33 secure' and 'Cortex-M33 non secure'. The 'Cortex-M33 non secure' option is selected. Below the mode selection, there are checkboxes for 'Activate Clock Source' and 'Activate Calendar', both of which are checked and highlighted with red boxes and red circles labeled '2' and '3' respectively. There are also dropdown menus for 'Alarm A', 'Alarm B', 'WakeUp', and 'Calibration', all set to 'Disable'. A checkbox for 'Timestamp' is checked and highlighted with a red box and a red circle labeled '4'. The 'Configuration' section shows a 'Reset Configuration' button and a list of settings: 'User Constants', 'Parameter Setting', 'RTC Security', 'NVMC Settings', 'GPIO Settings', and 'RTC Privilege'. The 'RTC Security' option is highlighted with a red box and a red circle labeled '5'. Below the 'RTC Security' option, there is a section 'Configure the below parameters :'. The 'Secure RTC' section is expanded, showing a table of parameters and their security status. The 'RTC full secure' parameter is set to 'Disable' and is highlighted with a red box and a red circle labeled '5'. The 'Features' section is also expanded, showing a table of features and their security status. The 'RTC full secure' parameter is set to 'Disable' and is highlighted with a red box and a red circle labeled '5'.

Component	M33S	M33NS
LPTIM1	☐	☐
LPTIM2	☐	☐
LPTIM3	☐	☐
LPTIM4	☐	☐
LPTIM5	☐	☐
LPTIM6	☐	☐
RTC	☐	☐
TAMP	☒	☐
TIM1	☐	☐
TIM2	☐	☐
TIM3	☐	☐
TIM4	☐	☐
TIM5	☐	☐
TIM6	☐	☐
TIM7	☐	☐
TIM8	☐	☐
TIM12	☐	☐
TIM13	☐	☐
TIM14	☐	☐
TIM15	☐	☐
TIM16	☐	☐
TIM17	☐	☐

Parameter	Value
RTC full secure	Disable

Feature	Value
RTC Initialisation	Non-Secure
RTC Alarm A	Non-Secure
RTC Alarm B	Non-Secure
RTC TimeStamp	Non-secure
RTC Calibration	Non-Secure
RTC WakeUpTimer	Non-Secure

Keep RTC on non secure side



Activate the RTC Interrupt

Clock Configuration

▼ Software Packs
▼ Pinout

RTC Mode and Configuration

Mode

Runtime contexts:

Cortex-M33 secure	Cortex-M33 non secure
☒	☐

- ☒ Activate Clock Source
- ☒ Activate Calendar
- Alarm A ▼
- Alarm B ▼
- ☒ Timestamp
- WakeUp ▼
- Calibration ▼
- ☐ Reference clock detection

Configuration

Reset Configuration

✔ User Constants

✔ NVIC Settings

✔ GPIO Settings

✔ Parameter Settings

✔ RTC Security

✔ RTC Privilege

NVIC_NS Interrupt Table	Enabled	Preemption Priority	Sub Priority
RTC non-secure interrupt	☒		0

RTC interrupt will trigger as soon as input event is detected on PC13



Activate the ICACHE and generate code

Pinout & Configuration | Clock Configuration | Project

Software Packs | Pinout

ICACHE M... Configuration

Runtime contexts:

Cortex-M33 secure ☐ Cortex-M33 non secure ☒

☐ Memory address remap

Mode 1-way (direct mapped cache)

Secure Attribute not secured

Reset Configuration

Parameter Settings | NVIC Settings

System Core

	M33S	M33NS
CORTEX_M33		
CORTEX_M33_S	☒	
DCACHE1	☐	☐
FLASH	☐	☐
GPDMA1	☒	☒
GPDMA2	☒	☒
ICACHE	☒	☐
NVIC_NS		☒
NVIC_S	☒	
RAMCFG	☐	☐
RCC	☒	☒
SAU	☒	
SYS_NS		☒
SYS_S	☒	
WWDG	☐	☐

GENERATE CODE

Project Manager | Tools

MX Code Generation

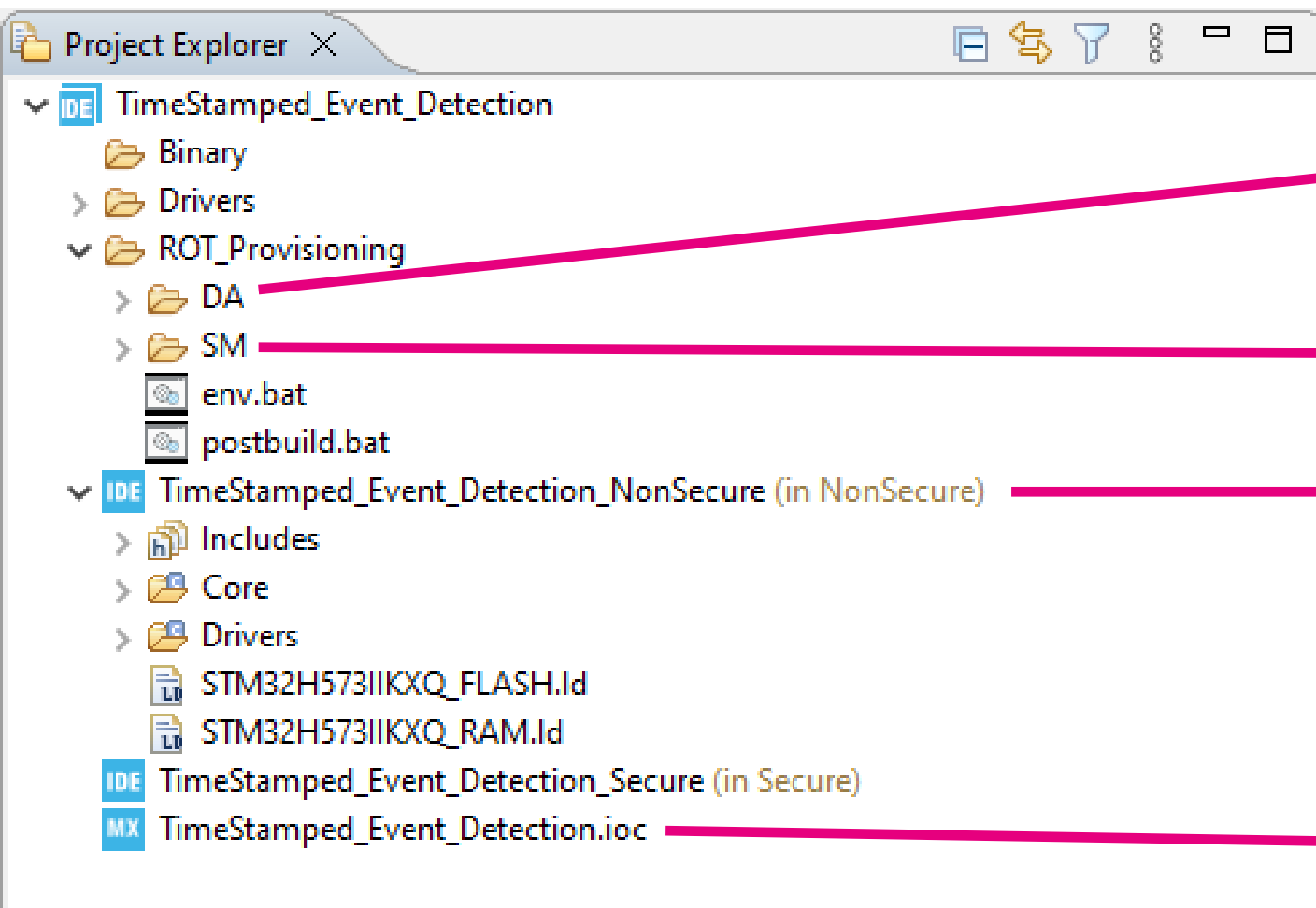
The Code is successfully generated under :

C:/Training/STM32H5..._Security_WS/Hands-on/TimeStamped_Event_Detection

Project language : C

Open Folder | Open Project | Close

Project generated



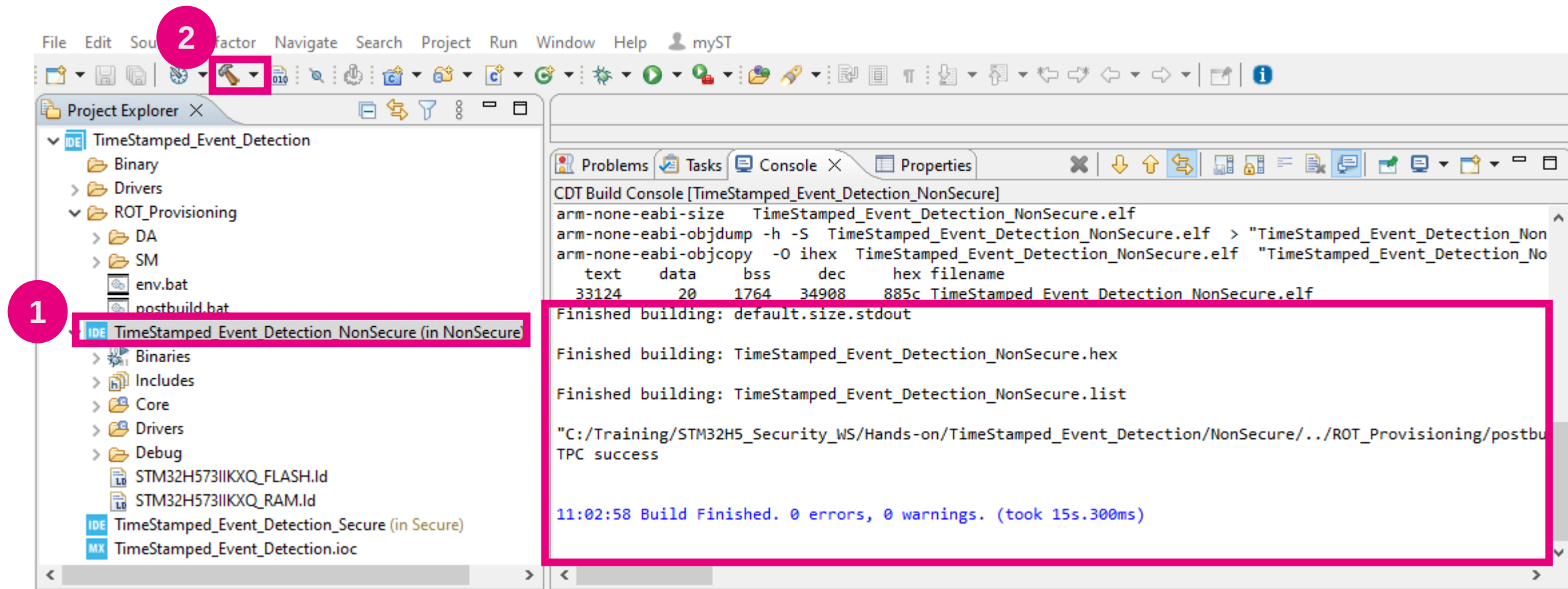
Debug Authentication : all the scripts and resource associated

Secure Manager : all the scripts and resource associated

User application code

User application ioc

Build the application



1

2

CDT Build Console [TimeStamped_Event_Detection_NonSecure]

```
arm-none-eabi-size TimeStamped_Event_Detection_NonSecure.elf
arm-none-eabi-objdump -h -S TimeStamped_Event_Detection_NonSecure.elf > "TimeStamped_Event_Detection_NonSecure.elf.map"
arm-none-eabi-objcopy -O ihex TimeStamped_Event_Detection_NonSecure.elf "TimeStamped_Event_Detection_NonSecure.elf.hex"
text      data      bss      dec      hex filename
33124      20      1764    34908    885c TimeStamped_Event_Detection_NonSecure.elf
```

Finished building: default.size.stdout

Finished building: TimeStamped_Event_Detection_NonSecure.hex

Finished building: TimeStamped_Event_Detection_NonSecure.list

"C:/Training/STM32H5_Security_WS/Hands-on/TimeStamped_Event_Detection/NonSecure/../../ROT_Provisioning/postbuild.bat" success

11:02:58 Build Finished. 0 errors, 0 warnings. (took 15s.300ms)

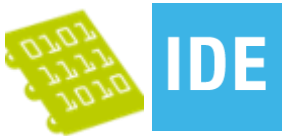
Build Analyzer Static Stack Analyzer Cyclomatic Complexity

TimeStamped_Event_Detection_NonSecure.elf - /TimeStamped_Event_Detection_NonSecure/Debug - Sep 20, 2023,

Memory Regions Memory Details

Region	Start address	End address	Size	Free	Used	Usage (%)
FLASH	0x0806e400	0x080ffbfe	582 KB	549.63 KB	32.37 KB	5.56%
RAM	0x20000000	0x2003bfff	240 KB	238.27 KB	1.73 KB	0.72%

Flash and RAM mapping automatically managed by Boot Path



Add source file code in the project

1 Drag and drop: detection.c and trace.c
In Core\Src folder

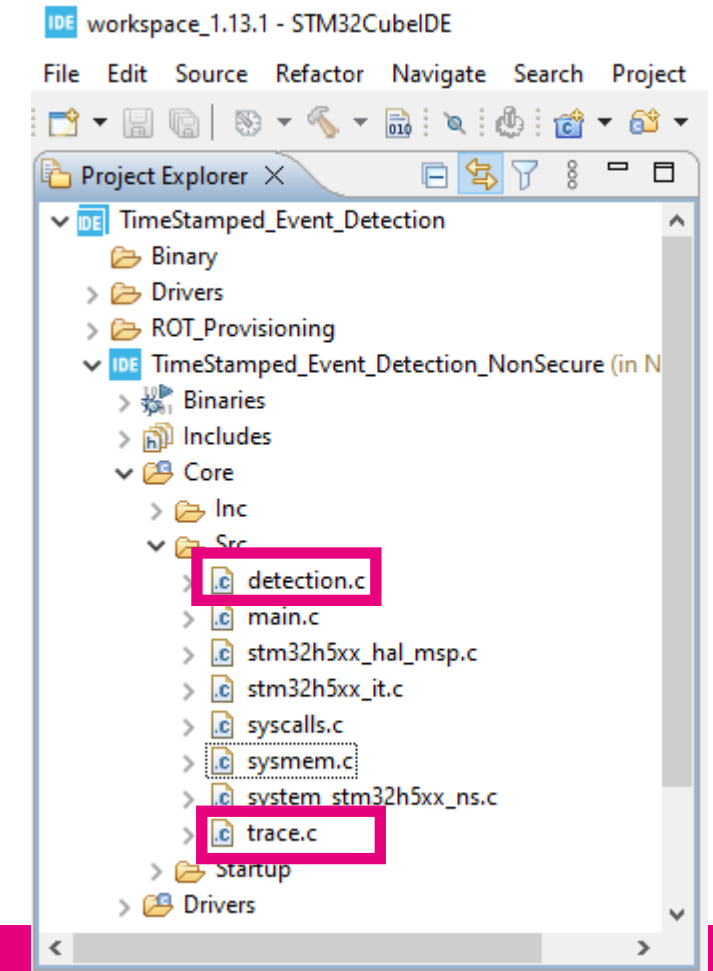
2 Select how files should be imported into the project:

- ☒ Copy files
- ☐ Link to files

☒ Create link locations relative to: PROJECT_LOC

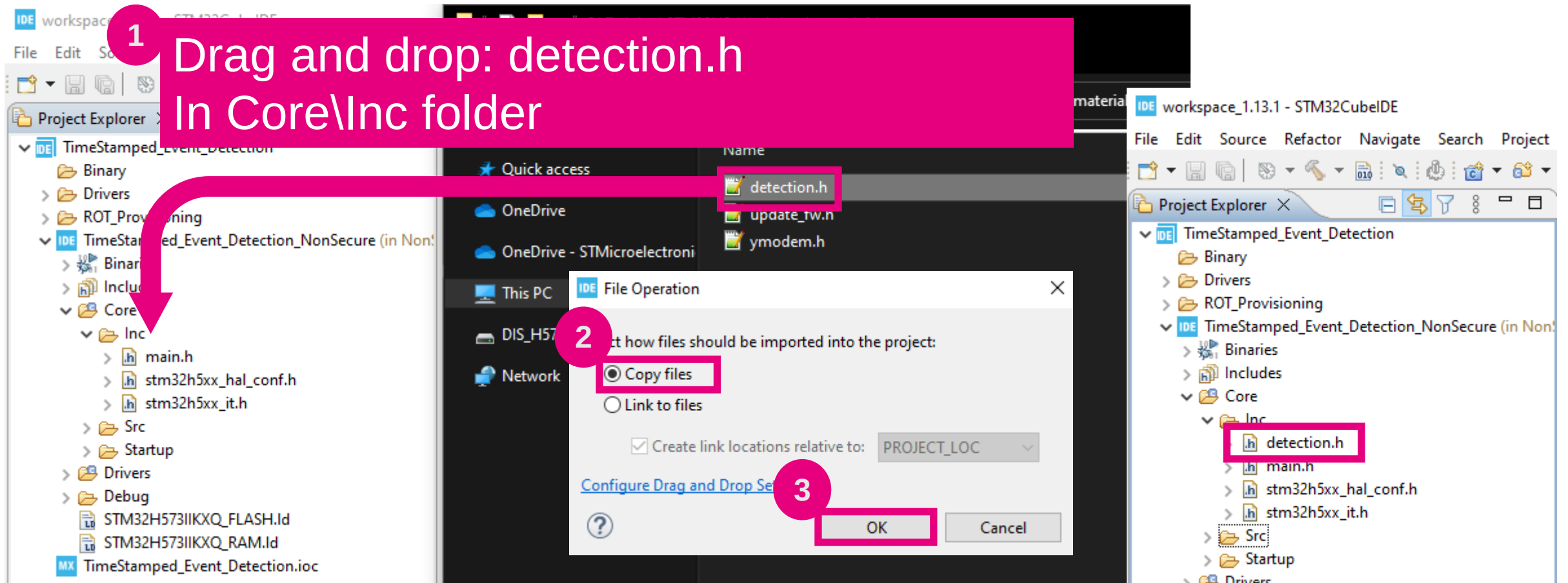
[Configure Drag and Drop Settings](#)

3 OK



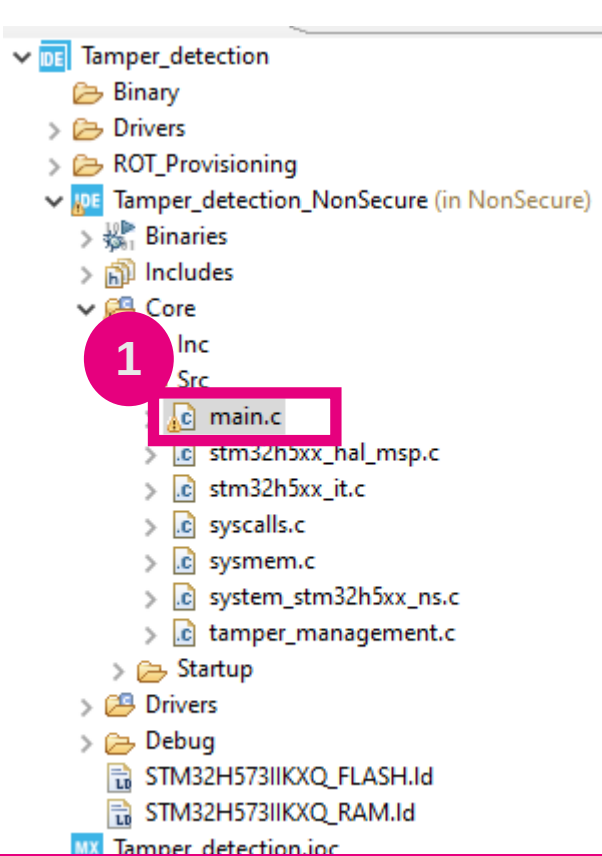
Material location :
C:\Training\STM32H5_Security_WS\Hands-on-material\
TimeStamped_Event_Detection\Project_patch_files\Src

Add header file code in the project



Material location :
C:\Training\STM32H5_Security_WS\Hands-on-material\
TimeStamped_Event_Detection\Project_patch_files\Inc

Modify the code



2

```

22 /* Private includes -----*/
23 /* USER CODE BEGIN Includes */
24 #include "detection.h"
25 /* USER CODE END Includes */
26

```

3

```

100 /* USER CODE BEGIN 2 */
101 printf("*****\r\n");
102 printf("TimeStamped Event detection version 1\r\n");
103 printf("*****\r\n");
104 /* USER CODE END 2 */

```

4

```

106 /* Infinite loop */
107 /* USER CODE BEGIN WHILE */
108 while (1)
109 {
110     ProcessEventDetection();
111     /* USER CODE END WHILE */
112
113     /* USER CODE BEGIN 3 */
114 }
115 /* USER CODE END 3 */

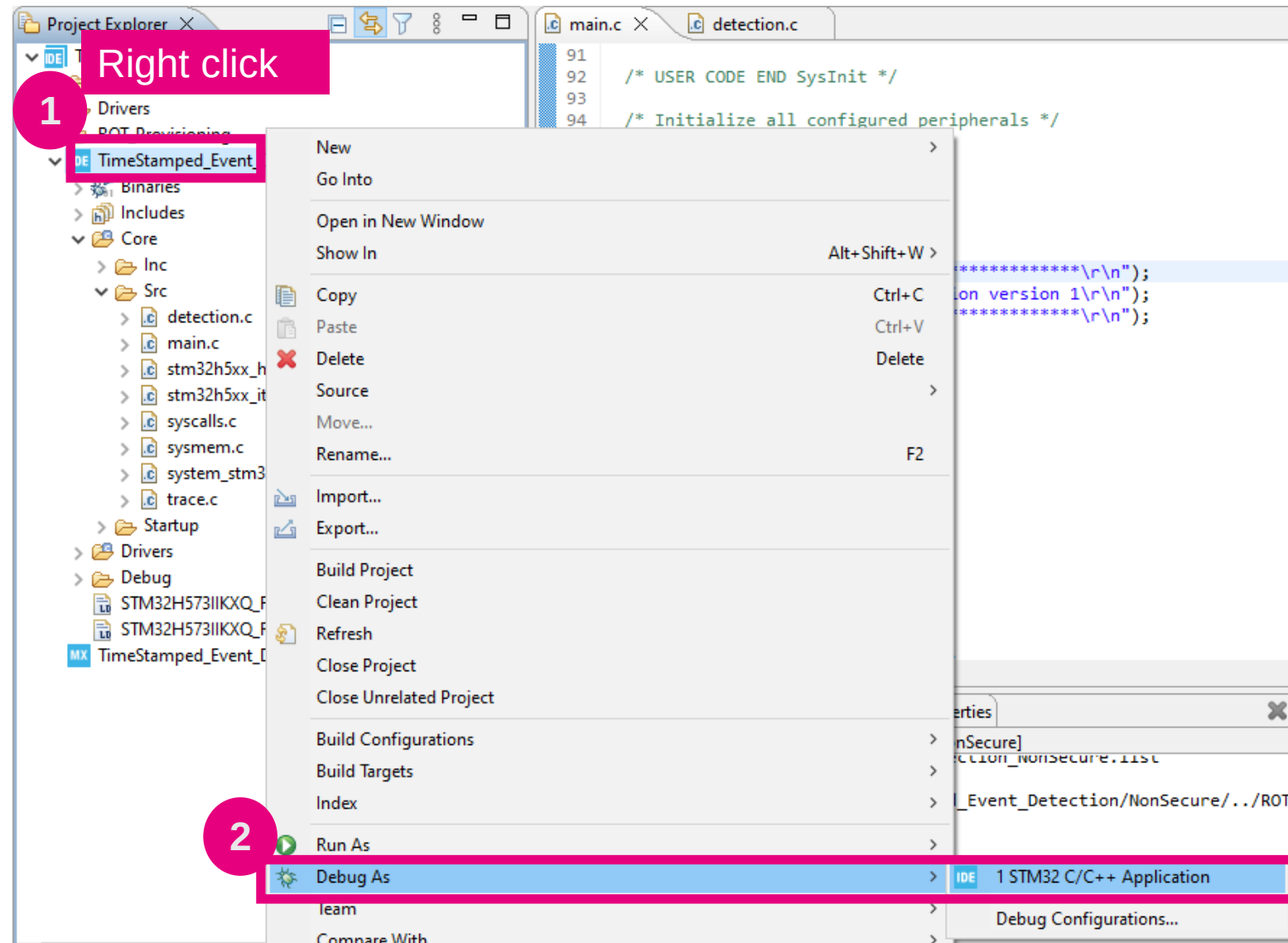
```

Material location :

C:\Training\STM32H5_Security_WS\Hands-on-material\

TimeStamped_Event_Detection\CodeToAdd.txt

Compile and launch the debug



Modify the debug configuration

IDE Edit Configuration

Edit launch configuration properties

1 **Debugger**

Name: **Test_Sensor_Event_Detection_NonSecure**

Mail: **Debugger** Startup Source Common

GDB Connection Settings

☒ Autostart local GDB server Host name or IP address: **localhost**

☐ Connect to remote GDB server Port number: **61234**

Debug probe: **ST-LINK (ST-LINK GDB server)**

GDB Server Command Line Options

Show Command Line

Interface

☒ SWD ☐ JTAG

☐ ST-LINK S/N **Scan**

Frequency (kHz): **Auto**

Access port: **1 - Cortex-M33**

Reset behaviour

Type: **Connect under reset**

2 **Connect under reset**

Software system reset

☐ Hardware reset

☐ Core reset

☐ None

Key path: **Browse...**

Certificate path: **Browse...**

Permissions: **Debug Non Secure L3**

Device settings

Debug in low power modes: **Enable**

3 **OK**

Revert Apply

Connect under reset is not possible. Debugger can only attach when target is running in non secure

IDE Confirm Perspective Switch

This kind of launch is configured to open the Debug perspective when it suspends.

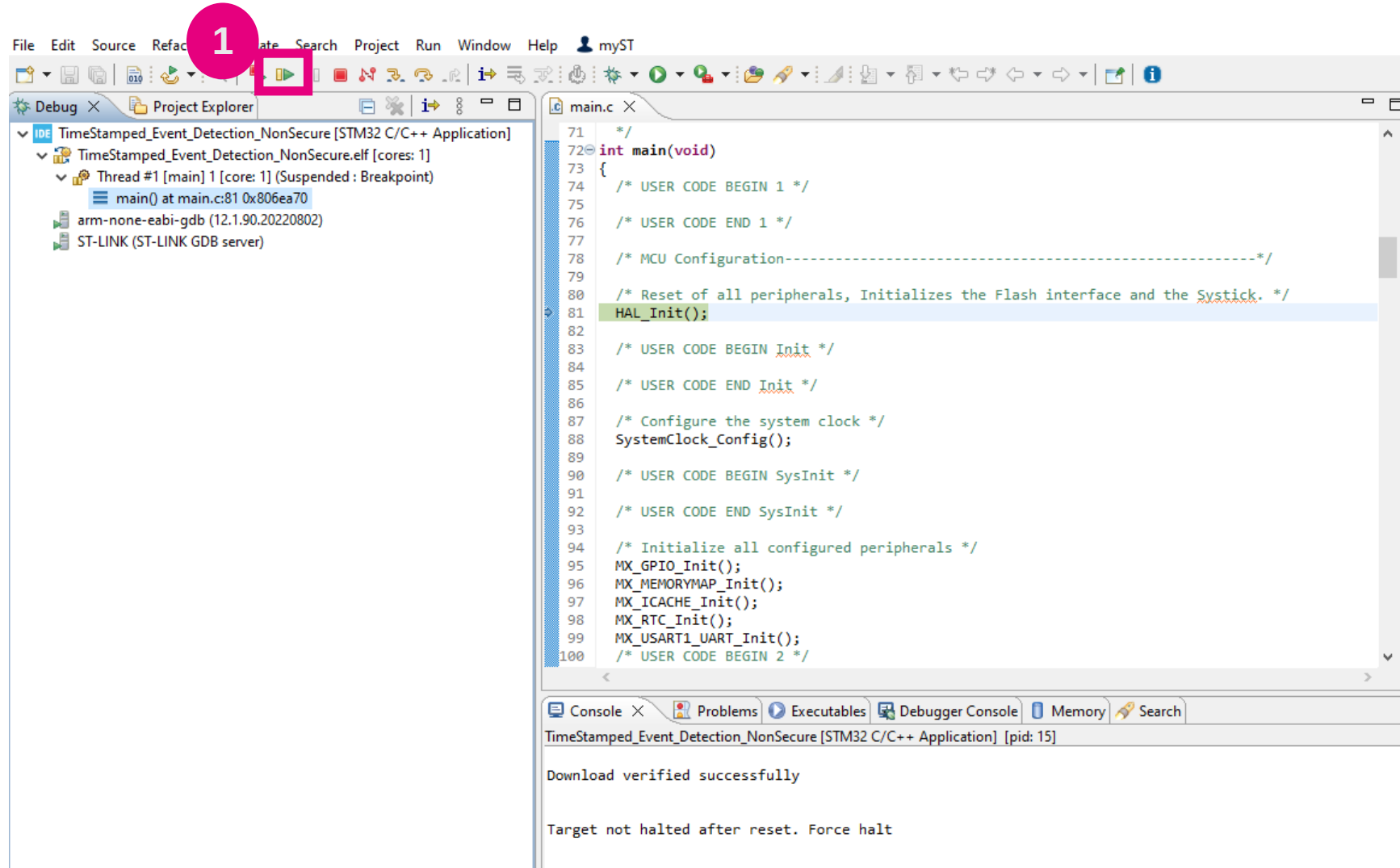
This Debug perspective supports application debugging by providing views for displaying the debug stack, variables and breakpoints.

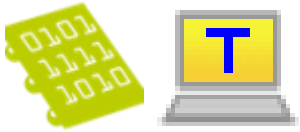
Switch to this perspective?

4 ☒ Remember my decision

5 **Switch** No

Debugger connected to target





Basic application is ready !

- Press blue user button to trigger the event

```
VT COM77 - Tera Term VT
File Edit Setup Control Window Help
*****
TimeStamped Event detection version 1
*****

*****Event detected*****
Time : 00:13:22
Date : 01-01-2023

*****Event detected*****
Time : 00:13:23
Date : 01-01-2023
█
```




Implementation details

```
void HAL_RTCEx_TimeStampEventCallback(RTC_HandleTypeDef *RTC_Handle)
{
    if ( g_EventDetected == 0 )
    {
        HAL_RTCEx_GetTimeStamp(&hrtc, &sTime, &sDate, RTC_FORMAT_BIN);
        g_EventDetected = 1;
    }
}
```

When you press button, this call back memorize the time and the date

```
void ProcessEventDetection()
{
    if (g_EventDetected)
    {
        char *event = BuildOutputEvent();
        OutputEventInClear(event);
#ifdef OUTPUT_ENC_DATA
        OutputEventInEncrypted(event);
#endif
        g_EventDetected=0;
    }
}
```

In this procedure, if an event as been detected :

- we format the output
- we send the output to USART

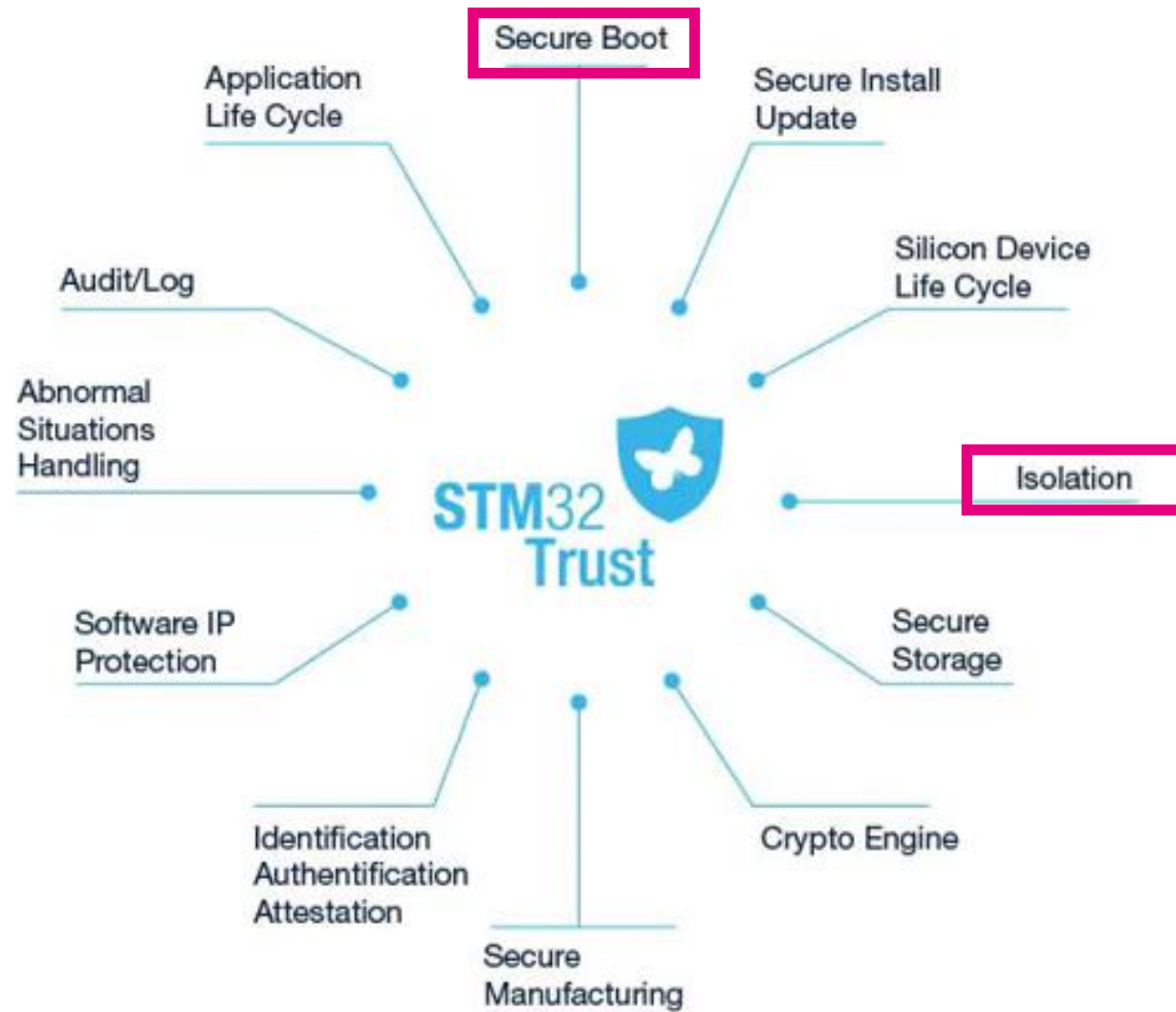


Conclusion

- During this first hands-on we created a basic application launch by the secure boot.
- STM32CubeMX implements the new Boot Path feature allowing easy secure boot integration
- STM32CubeMX provides easy integrate application code, with adapted linker file on of the Secure Manager, creating the non secure
- You could also see how the platform boots in secure manager context



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